



MODULE · PROCEDURE

VIRTUAL CONTROL

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SDDC Manager Credential Reconnect

Recovering the vCenter service account and NSX API access after an out-of-band vCenter redeploy.

v1.0

VCF 9.0.1 · SDDC Manager · NSX 9.0.1

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INTERNAL

SDDC Manager Credential Reconnect

Prepared by: Virtual Control LLC **Date:** June 13, 2026 **Document Version:** 1.0 **Classification:** Internal

What this fixes. After a vCenter Server is **redeployed out of band** (rebuilt rather than restored), SDDC Manager loses the ability to authenticate to vCenter, and — by extension — to validate or remediate **any** NSX credential. Password Management shows accounts **Disconnected**, credential operations fail with `vim.fault.InvalidLogin`, and inventory collection stalls. This module restores the management plane end-to-end with **no support contract and no database surgery**.

Two independent fixes, applied in order: 1. **Recreate SDDC Manager's vCenter service account** in the redeployed vCenter's SSO (it does not exist in the fresh SSO). 2. **Clear the NSX API authentication lockout** that the prior failed attempts triggered.

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1. Summary

A vCenter redeploy stands up a **fresh SSO domain**. SDDC Manager authenticates to vCenter using a service account it created during bring-up — `svc-sddc-manager-vcenter-<id>@vsphere.local` — and that account exists only in the *old* SSO. The redeployed vCenter has **no such user**, so every login fails.

Because SDDC Manager's credential subsystem validates NSX accounts **through the vCenter session**, the same `InvalidLogin` surfaces on NSX credential operations — which is why the symptom is easily misread as an "NSX SSH problem." It is not. Fix the vCenter service account and the NSX symptom changes; the only residual is an NSX **API auth-policy lockout** left over from the earlier failed retries, which is cleared separately.

Outcome when complete: every entry in **Administration** → **Password Management** reads **Active** — the vCenter and NSX service accounts, every ESXi `svc-vcf-*` account, and NSX `root / admin / audit` — and vCenter inventory resumes collecting.

2. Symptoms

Where	What you see
Password Management	vCenter and/or NSX accounts Disconnected ; Remediate/Update fail
Credential operation	Failed to ... NSX user details / <code>vim.fault.InvalidLogin</code> : Cannot complete login due to an incorrect user name or password
<code>operationsmanager.log</code>	repeated <code>Remove session ... "username":"svc-sddc-manager-vcenter-<id>@vsphere.local"</code> then <code>vim.fault.InvalidLogin</code> ; <code>Query with errors / DataService call failed</code> ; Data service result obtained for 0 hosts
After Fix 1, NSX <code>root</code> only	Failed to retrieve NSX user details because the password is not in sync (an NSX HTTP 403 — the API lockout, not a real mismatch)

3. Root cause

- **Primary:** the SDDC Manager vCenter service account is **absent from the redeployed vCenter's SSO**. Confirmed with `dir-cli user find-by-name` → `ERROR_NO_SUCH_USER (1317)`.
- **Why it masquerades as NSX:** the NSX credential validation rides the vCenter session, so a dead vCenter session throws `vim.fault.InvalidLogin` on NSX operations too.
- **Residual:** NSX's API auth-policy locks an account after `max-auth-failures` (default **5**) for `lockout-period` (default **900 s**). The earlier failed remediate tripped it, and **every Retry re-arms the timer**, so it never self-clears.

Ruled out first (verify, don't guess): SDDC Manager's `known_hosts` entries for the rebuilt NSX were already **current** (host-key theory disproved by direct fingerprint comparison); NSX `root` was **not locked**, **not expired**, password correct (the appliance accepted SSH and the NSX API answered from localhost). The failure was upstream, in the vCenter session — proven by reading the stack trace down to the failing caller.

4. Prerequisites

Item	Value / note
vCenter	Reachable; SSO administrator credential known (administrator@vsphere.local)
vCenter root SSH	Enabled; note root SSH lands in appliance sh — type shell for bash
SDDC Manager shell	vcf user; the API-retrieval steps do not require root
NSX admin	nsxcli access (ssh admin@nsx-vip...) for the lockout step
Change window	Yes — Fix 1 writes an SSO account + permission; Fix 2 toggles an NSX security policy
Secrets	All passwords below are <vault> references — never store literal values in this document

5. Procedure

5.1 Diagnose — confirm it is the vCenter service account

On SDDC Manager (vcf , then su - to read the protected log), find the failing login:

```
L=/var/log/vmware/vcf/operationsmanager/operationsmanager.log
grep -nE "^20[0-9-]+T.*(InvalidLogin|Query with errors|DataService call failed)" "$L" | tail -3
```

Look for the session being removed for svc-sddc-manager-vcenter-<id>@vsphere.local against https://<vcenter-fqdn>:443/sdk with Cannot complete login due to an incorrect user name or password.

On vCenter (ssh root@<vcenter-fqdn> → shell), confirm the account is missing:

```
/usr/lib/vmware-vmafd/bin/dir-cli user find-by-name \
--account svc-sddc-manager-vcenter-<id> \
--login administrator@vsphere.local
# expected: dir-cli failed. Error 1317: ERROR_NO_SUCH_USER
```

The <id> suffix is deployment-specific. Take the exact account name from the username field returned in step 5.2.

5.2 Fix 1 — recreate the vCenter service account (KB 402863)

Step A — retrieve the password SDDC Manager already stores (KB 327195). On SDDC Manager:

```
TOKEN=$(curl -s -d '{"username":"administrator@vsphere.local","password":"<sso-admin-password>"}' \
-H "Content-Type: application/json" -X POST http://127.0.0.1/v1/tokens | jq -r '.accessToken')
```

```
curl -s -k -H "Authorization: Bearer $TOKEN" \
'https://localhost/v1/system/credentials/service' | json_pp | grep -iA6 -B2 vcenter
```

Note the block with `"targetType":"VCENTER"`, `"credentialType":"SSO"` — record its `username` (the exact account name) and `secret` (the stored password).

The entire JSON payload is single-quoted so the shell does not mangle special characters in the password.

Step B — recreate the account in vCenter SSO with that exact secret. On vCenter (`shell`). Each command prompts for the SSO administrator password; leaving it off the command line keeps it out of shell history.

```
DIR=/usr/lib/vmware-vmafd/bin/dir-cli
ADMIN=administrator@vsphere.local

$DIR user create --account svc-sddc-manager-vcenter-<id> \
  --user-password '<service-account-secret>' \
  --first-name svc-sddc-manager-vcenter-<id> --last-name svc --login $ADMIN

$DIR user modify --account svc-sddc-manager-vcenter-<id> --password-never-expires --login $ADMIN

$DIR group modify --name Administrators --add svc-sddc-manager-vcenter-<id> --login $ADMIN

$DIR user find-by-name --account svc-sddc-manager-vcenter-<id> --level 2 --login $ADMIN
```

Why these three: using the **stored** secret means SDDC Manager's database already matches — no rotation needed. `--password-never-expires` prevents the SSO password policy from expiring it (and avoids the far-future-expiry edge case in KB 381321). `Administrators` (of `vsphere.local`) carries the vCenter Administrator role SDDC Manager requires. The final `find-by-name` should now report the account **enabled, not locked, password never expires**.

Step C — verify. On SDDC Manager, within ~1–2 minutes:

```
date -u
grep -nE "^20[0-9-]+T.*(InvalidLogin|DataService call failed)" "$L" | tail -2 # should predate the
recreate
grep -n "Data service result obtained for" "$L" | tail -3 # should show >0 hosts
```

The `InvalidLogin` flood stops; queries return data; **Password Management shows the vCenter service account Active**.

5.3 Fix 2 — clear the NSX API auth-lockout (KB 314647)

After Fix 1, an NSX `root` Update may report **"Failed to retrieve NSX user details because the password is not in sync."** That string is an NSX **HTTP 403** (`module_name: common-services`, "credentials incorrect or account locked") — the API lockout, not a real mismatch.

Confirm the policy (nsxcli — `ssh admin@nsx-vip...`):

```
get auth-policy api lockout-period
get auth-policy api lockout-reset-period
get auth-policy api max-auth-failures
```

Clear it cleanly:

1. In SDDC Manager Password Management, **Cancel** the pending NSX task. **Do not click Retry** — each retry re-arms the 900-second timer.
2. Temporarily lift the lockout (nsxcli): `set auth-policy api lockout-period 0`
3. Run **one** Remediate for NSX `root` in SDDC Manager (password `<vault>`).
4. Confirm `root` flips to **Active**, then **restore the policy**: `set auth-policy api lockout-period 900`

If multiple NSX accounts are affected, the same single-attempt discipline applies to each. The OS-level lock (`/usr/sbin/faillock --user <u> --reset`) is separate from this API lockout and is usually clear — check only if SSH/console login itself is refused.

5.4 Final verification

Check	Expected
Password Management	All Active — vCenter + NSX service accounts, 4× ESXi, NSX <code>root</code> / <code>admin</code> / <code>audit</code>
operationsmanager.log	no new <code>InvalidLogin</code> / <code>DataService call failed</code> ; Data service result obtained for N hosts (N > 0)
vCenter datasource / inventory	collecting; all hosts present

6. Notes & cautions

- **Read the fault's origin, not its label.** `vim.fault.InvalidLogin` on an NSX operation was the *vCenter* service account failing upstream. Follow the stack trace to the failing caller before naming a root cause.
- **Stop retrying a locked NSX account** — every Retry resets the lockout window. Cancel, lift the policy, fire one attempt, restore.
- **Administrators is the assumption to be aware of.** Broadcom's KB does not enumerate the exact group; `Administrators` grants the full vCenter admin the service account needs. Fully reversible: `group modify --name Administrators --remove ... / user delete`.
- **Single-quote the stored secret** in `dir-cli` — it commonly contains shell metacharacters (`*` , `|` , `%` , `=`).
- **vCenter root SSH opens appliancesh**, not bash — type `shell` first.
- **No DB surgery, no snapshots.** This procedure touches only the SSO account and the NSX auth policy; both are reversible.

7. References

KB / source	Topic
KB 402863	Recreate the missing SDDC Manager service account in vCenter
KB 327195	Retrieve SDDC Manager service-account credentials (/v1/system/credentials/service)
KB 399460 / 381321	vCenter SDDC service accounts disconnected — reset/match procedure
KB 314647	NSX-T credential ops fail with 403 "credentials incorrect or account locked" (API lockout)
KB 314657	Credential operations fail on NSX-T components
KB 439290	NSX root password disconnected / remediation fails — Update vs Remediate
KB 408913	NSX certificate trust in SDDC Manager (certificate_unknown(46)) — prerequisite
KB 316028	SDDC Manager stale SSH host keys (known_hosts refresh) — prerequisite

8. Document Information

Field	Value
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Distribution	Lab notebook

Revision History

Version	Date	Notes
1.0	2026-06-13	Initial release. Written from the live remediation: recreate the missing vCenter SSO service account with the SDDC-stored secret (KB 402863/327195), then clear the NSX API auth-lockout (KB 314647). Captures the diagnosis that distinguishes the upstream vCenter-session failure from an apparent NSX-SSH problem.